

CASCADING STYLE SHEET (CSS)



CSS is a simple mechanism of describing common presentation semantics for every page in a website. It can control the layout of multiple web pages all at once.

Version of CSS:

CSS 1

CSS 2

CSS 3

CSS SYNTAX

Selector { Property1: property1-value; Property2: property2-value; PropertyN: propertyN-value; }

Selector {
 Property1: property1-value;
 Property2: property2-value;
 PropertyN: propertyN-value;
}

Selector - The selector points to the HTML element or HTML Tag you want to style. It is used to find or select HTML elements based on their element name, id, class, attribute, and more.

Property – It indicates that these properties are defined for that element or selector.

Property-Value – These are values assigned to Properties.

Syntax: -

Selector { Property1: property1-value; Property2: property2-value; PropertyN: propertyN-value; }

Ex: -

```
p { color: red; font-size: 22px;}
```

```
p {  
    color: red;  
    font-size: 22px;  
}
```

ELEMENT SELECTORS

The element selector selects elements based on the element name.

Syntax: -

Selector { Property1: property1-value; Property2: property2-value; PropertyN: propertyN-value; }

Ex: -

```
p { color: red; font-size: 22px;}
```

```
h1 {color: blue; font-size: 22em;}
```

WAY OF INSERTING CSS

- ▶ External style sheet
- ▶ Internal style sheet/ Embedded Style Sheet
- ▶ Inline style

EXTERNAL STYLE SHEET

An external style sheet is a separate document that contains only CSS rules. An external style sheet helps to change the look of an entire website by changing just one css file. It should not contain any HTML Tags. It has .css extension.

EXAMPLE

```
p { color: red; font-size: 24px;}  
h1{ color: blue; font-size: 24em;}
```

1. Open notepad++ or any other Editor or IDE
2. Write CSS Code
3. Save with .css extension for example yasham.css

HOW TO LINK WEB PAGE TO AN EXTERNAL STYLE SHEET

The href attribute with <link> element inside the <head> tag is used to link web page to an external style sheet.

```
<html>
  <head>
    <title>Welcome to Yasham</title>
    <link rel="stylesheet" href="yasham.css">
  </head>
  <body>
    <h1>I am Heading</h1>
    <p>I am first Paragraph.</p>
    <p>I am second Paragraph</p>
  </body>
</html>
```

INTERNAL STYLE SHEET

Internal Style sheet is a set of style that is created as a part of HTML document. An internal style sheet may be used if one single page has a unique style.

Internal Style sheets are created using `<style>` element, which is added inside the `<head>` element of the HTML document.

```
<head>
  <title>Hello CSS</title>
  <style type="text/css">
    p { color: red; font-size: 24px;}
    h1{ color: blue; font-size: 24em;}
  </style>
</head>
<body>
<h1>I am Heading</h1>
<p>I am first Paragraph.</p>
<p>I am second Paragraph</p>
</body>
</html>
```

INLINE STYLES

Inline style is useful when we need to define specific style for individual elements present on a web page. The *style* attribute in a specific Tag or element, is used to create inline style. The style attribute can contain any CSS property between double quotes.

Ex:-

For paragraph:-

```
<p style="color: red; font-size: 22px;"> I am first paragraph</p>
```

```
<p> I am second paragraph</p>
```

For Heading: -

```
<h1 style="color: red; font-size: 22px;"> I am Heading</h1>
```

```
h1{ color: orange; font-size: 25em;}  
p { color: green; font-size: 20px;}
```

Save as : yasham.css

MULTIPLE STYLE SHEET



```
<html>
  <head>
    <title>Hello CSS</title>
    <link rel="stylesheet" href="yasham.css">
    <style type="text/css">
      h1{color: blue; font-size: 35em;}
      p {color: red; font-size: 30px;}
    </style>
  </head>
  <body>
    <h1 style="color: cyan font-size: 45em;" >I am Heading</h1>
    <p style="color: yellow font-size: 40px;" >I am 1st Para</p>
  </body>
</html>
```

External Style Sheet

Internal Style Sheet

Inline Styles

PRIORITY OF STYLE SHEETS

- ▶ Inline Styles
- ▶ External or Internal Style Sheets
- ▶ Browser default

PRIORITY - EXTERNAL OR INTERNAL

If the internal style is defined after the link to the external style sheet then Internal Style has highest priority.

Ex: -

```
<html>
  <head>
    <title>Hello CSS</title>
    <link rel="stylesheet" href="yasham.css">
    <style type="text/css">
      p { color: red; font-size: 30px;}
      h1 { color: blue; font-size: 35em;}
    </style>
  </head>
```

External Style Sheet

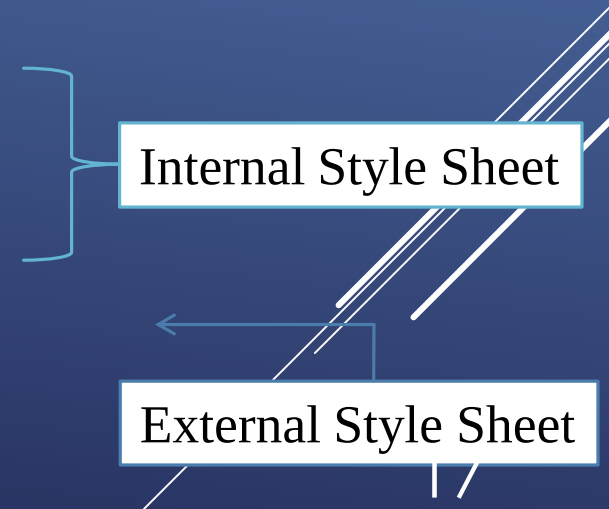
Internal Style Sheet

PRIORITY - EXTERNAL OR INTERNAL

If the internal style is defined before the link to the external style sheet then External Style has highest priority.

Ex: -

```
<html>
  <head>
    <title>Hello CSS</title>
    <style type="text/css">
      p { color: red; font-size: 30px;}
      h1 { color: blue; font-size: 35em;}
    </style>
    <link rel="stylesheet" href="yasham.css">
  </head>
```



PRIORITY OF STYLE SHEETS



- ▶ **Inline Styles**
- ▶ **External or Internal Style Sheets**
- ▶ **Browser default**

COMMENT

A CSS comment starts with /* and ends with */.

```
<html>
  <head>
    <style>
      #yasham {
        /* I am comment */
        color: red; font-size: 60px;
      }
    </style>
  </head>
  <body>
    <p id="yasham">Hello World!</p>
    <p>This paragraph is not affected by the style.</p>
  </body>
</html>
```

ID SELECTOR

The id selector uses the id attribute of an HTML element to select a specific element.

The id of an element should be unique within a page, so the id selector is used to select one unique element.

To select an element with a specific id, write a hash (#) character, followed by the id of the element.

RULES

- ▶ Must begin with a letter A-Z or a-z
- ▶ Id name cannot start with a number
- ▶ Must not contain any space characters
- ▶ Can be followed by: letters (A-Za-z), digits (0-9), hyphens ("-"), and underscores ("_")
- ▶ In HTML, all values are case-insensitive

```
<html>
  <head>
    <style>
      #yasham {
        color: red; font-size: 60px;
      }
    </style>
  </head>
  <body>
    <p id="yasham">Hello World!</p>
    <p>This paragraph is not affected by the style.</p>
  </body>
</html>
```

CLASS SELECTOR/ STYLE CLASS



Class selectors are used where you want to apply a style either on some of tag or across the several tags without repeating the style rule in an HTML document. Using this method of creating styles, you can create styles in the form of style classes in an external style sheet or as an internal style sheet. The class attribute is used to apply style class in HTML Tag.

TYPE OF STYLE CLASS

- ▶ Universal style classes
- ▶ Element specific style classes

RULES

- ▶ Must begin with a letter A-Z or a-z
- ▶ A class name cannot start with a number
- ▶ Must not contain any space characters
- ▶ Can be followed by: letters (A-Za-z), digits (0-9), hyphens ("-"), and underscores ("_")
- ▶ In HTML, all values are case-insensitive

UNIVERSAL STYLE CLASSES

Universal style class starts with a dot operator followed by the class name.

Syntax: -

```
<style>
```

```
  .class_name { class definition }
```

```
</style>
```

Ex:-

```
<style>
```

```
  .red {color: #FF0000; font-size: 60px;}
```

```
</style>
```

HOW TO USE

```
<html>
  <head>
    <title>Yasham</title>
    <style>
      .red {color: #FF0000; font-size: 60px;}
    </style>
  </head>
  <body>
    <h2 class="red">I am Heading</h2>
    <p class ="red">I am first paragraph</p>
    <p>I am second paragraph</p>
  </body>
</html>
```

ELEMENT SPECIFIC STYLE CLASSES

An element specific style class starts with the element name, followed by a dot operator, which is followed by the class name.

Syntax: -

`<style>`

`element_name.class_name{class definition}`

`</style>`

Ex:-

`<style>`

`p.red {color: #FF0000; font-size: 60px;}`

`</style>`

HOW TO USE

```
<html>
  <head>
    <title>Yasham</title>
    <style>
      p.red {color: #FF0000; font-size: 60px;}
    </style>
  </head>
  <body>
    <h2 class="red">I am Heading</h2>
    <p class ="red">I am first paragraph</p>
    <p>I am second paragraph</p>
  </body>
</html>
```

USE TWO OR MORE CLASS

```
<html>
  <head>
    <title>Yasham</title>
    <style>
      p.red {color: #FF0000;}
      p.look{font-size: 60px;}
      .algn{text-align: center;}
    </style>
  </head>
  <body>
    <h2 class="algn">I am Heading</h2>
    <p class ="red look">I am first paragraph</p>
    <p class="red algn">I am second paragraph</p>
  </body>
</html>
```

GROUPING SELECTORS

To group selectors, separate each selector with a comma.

```
h1 {  
  color: red; font-size: 30px;  
}
```

```
p {  
  color: red; font-size: 30px;  
}
```

```
h2 {  
  color: red; font-size: 30px;  
}
```

```
h3 {  
  color: red; font-size: 30px;  
}
```

```
h1, p, h2, h3 {  
  color: red; font-size: 30px;  
}
```

color: red; font-size: 30px;

BACKGROUND

We can use background properties to enhance the background styles of the elements present in the HTML web page.

background-color
background-image
background-repeat
background-position
background-attachment

background-clip
background-size
background-origin
background

BACKGROUND-COLOR

The background-color property allows to set the background color of an HTML element. We can set this property to a predefined color name, color value or transparent.

Color name – *red, green, blue* etc

Color value

Hex value - *#F0E68C*

RGB value – *rgb(240,230,140)*

- ▶ `body { background-color: khaki; }`
- ▶ `body { background-color: #F0E68C; }`
- ▶ `body { background-color: rgb(240,230,140); }`

BACKGROUND-IMAGE

It is used to set background image of an HTML element. By default, the background-image property repeats an image both horizontally and vertically so it covers the entire element. While setting the background image for an HTML element, we should also specify a background color that will be used if the image is not available. We can set this property to *none* or url of an image.

- ▶ `body{background-image: url(image/geek.png);}`
- ▶ `body{background-image: url(geek.png);}`

BACKGROUND-REPEAT

It specifies whether the background image is repeated or not. We can set this property to *repeat*, *repeat-x*, *repeat-y* or *no-repeat*. By default, the background-image property repeats an image both horizontally and vertically so it covers the entire element.

Horizontal – repeat-x

Vertical – repeat-y

No repeat – no-repeat

```
body{  
    background-image: url(image/geek.jpg);  
    background-repeat: repeat-x;  
}
```

BACKGROUND-POSITION

It specifies the initial position of a background image. We can set this property to *Left top*, *Center top*, *right top*, *center left*, *center center*, *center right*, *left bottom*, *center bottom*, *right bottom*, *x% y%* , *x-position y-position*.

```
body{  
    background-image: url(image/geek.jpg);  
    background-repeat: no-repeat;  
    background-position: right top;  
}
```

BACKGROUND-ATTACHMENT



It specifies whether a background image is fixed or scrolls when the user scrolls the rest of the page. We can set this property to either scroll or *fixed*.

```
body{  
    background-image: url(image/geek.jpg);  
    background-repeat: no-repeat;  
    background-position: right top;  
    background-attachment: fixed;  
}
```

BACKGROUND

It works as a shorthand property for all background properties such as *background-color*, *background-image*, *background-repeat*, *background-position* and *background-attachment* etc. It sets all these background properties in one declaration.

```
body {background: khaki url(image/geek.jpg) no-repeat}
```

BACKGROUND-SIZE

The background-size property allows to specify the size of background image. The size can be specified in lengths, percentages, or by using one of the two keywords: *contain* or *cover*.

- ▶ contain - scales the background image to be as large as possible
- ▶ cover - scales the background image so that the content area is completely covered by the background image

```
body {  
    background-image: url(geek.jpg);  
    background-size: 80px;  
    background-repeat: no-repeat;  
}
```

```
body {  
    background-image: url(geek.jpg);  
    background-size: cover;  
    background-repeat: no-repeat;  
}
```

BACKGROUND-ORIGIN

The background-origin property specifies where the background image is positioned. We can set this property to *border-box*, *padding-box*, *content-box*.

- ▶ border-box - The background image starts from the upper left corner of the border.
- ▶ padding-box - The background image starts from the upper left corner of the padding edge. Its default value.
- ▶ content-box - The background image starts from the upper left corner of the content.

BACKGROUND-ORIGIN

```
p {  
  border: 5px solid black;  
  padding: 40px;  
  background-image: url(geek1.jpg);  
  background-size: 80px;  
  background-repeat: no-repeat;  
  background-origin: border-box;  
}
```

BACKGROUND-CLIP

The background-clip property specifies the painting area of the background.

We can set this property to *border-box*, *padding-box*, *content-box*.

- ▶ border-box - The background is painted to the outside edge of the border. Its default value
- ▶ padding-box - The background is painted to the outside edge of the padding.
- ▶ content-box - The background is painted within the content box.

```
body {  
  border: 5px dotted black;  
  padding: 40px;  
  background-color: khaki;  
  background-clip: content-box;  
}
```

MULTIPLE BACKGROUND IMAGE

```
body {  
    background-image: url(geek.jpg), url(geeky.jpg);  
    background-position: left top, right top;  
    background-size: 80px, 60px;  
    background-repeat: no-repeat, no-repeat;  
}
```

```
body { background: url(geek.png) left top no-repeat, url(geeky.png) right top no-repeat;  
}
```

FULL BACKGROUND IMAGE

html

```
{  
    background: url(bimage.jpg) no-repeat center fixed;  
    background-size: cover;  
}
```

TEXT

Text Properties provides various text formatting options.

- color
- direction
- text-align
- letter-spacing
- line-height
- text-decoration
- text-indent
- text-shadow
- text-transform
- vertical-align
- white-space
- word-spacing
- text-align-last
- text-overflow
- word-break
- word-wrap

COLOR

Color property set the color of text. We can set this property to either a color name or color value.

Color name – *red, green, blue* etc

Color value

Hex value - *#F0E68C*

RGB value – *rgb(240,230,140)*

```
h1{color: red;}
```

```
p{color: #F0E68C;}
```

DIRECTION

It specifies the text direction. We can set this property to either *ltr* (left to right) or *rtl* (right to left).

```
h1{direction: rtl;}
```

TEXT-ALIGN

It specifies alignment of text. We can set this property to *left*, *right*, *center*, or *justify*.

```
h1{text-align: right;}
```

```
p{text-align: center;}
```

LETTER-SPACING

It specifies the space between text characters. We can set this property to either normal or specify it in the form of the distance between two consecutive letters using identifier, such as px (pixels), in (inches), pt(points) and cm (centimeters).

Length = px, in, pt, cm, em

```
h1{letter-spacing: 50px;}
```

```
h1{letter-spacing: 3px}
```

LINE-HEIGHT

It specifies the distance between two lines. We can set this property to *normal* or specify it in the form of length, number or percentage.

```
p{line-height: 60%;}
```

```
h1{line-height: 150px;}
```

TEXT-DECORATION

It specifies the decoration added to the text. We can set this property to *none*, *underline*, *overline*, *line-through*.

```
h1{text-decoration: underline;}
```

TEXT-INDENT

It specifies the indentation of the first line of text in a block. we can specify this property in the form of either length or percentage.

Negative values are allowed. The first line will be indented to the left if the value is negative.

```
p{text-indent: 30px;}
```

TEXT-SHADOW

It specifies the shadow effect added to text. We can add more than one shadow to the text, separated by comma.

Syntax: - text-shadow: h-shadow v-shadow blur-radius color

h-shadow - The position of the horizontal shadow. Negative values are allowed.

v-shadow - The position of the vertical shadow. Negative values are allowed.

blur-radius - The blur radius. Default value is 0

color – The color of the shadow

none - No shadow

```
h1{text-shadow: 2px 2px 5px red;}
```

```
h2{text-shadow: 2px 2px 5px red, 5px 5px 8px blue;}
```

TEXT-TRANSFORM

text-transform is used to turn everything into uppercase or lowercase letters, or capitalize the first letter of each word. We can set this property to *none*, *uppercase*, *lowercase* or *capitalize*.

```
h1{text-transform: uppercase};
```

VERTICAL-ALIGN

It specifies the vertical positioning of text in the element. We can set this property to *baseline*, *sub*, *super*, *top*, *text-top*, *middle*, *bottom*, *text-bottom* or specify in the form length or percentage value.

```
.sp{vertical-align: super};
```

WHITE-SPACE

It specifies how white space inside an HTML element is handled. We can set this property to *normal*, *pre*, *nowrap*, *pre-line* or *pre-wrap*.

normal - Sequences of whitespace will collapse into a single whitespace. Text will wrap when necessary.

pre - Whitespace is preserved by the browser. Text will only wrap on line breaks. It is just like the `<pre>` tag in HTML

nowrap - Sequences of whitespace will collapse into a single whitespace. Text will never wrap to the next line. The text continues on the same line until a `<br>` tag is encountered

pre-line - Sequences of whitespace will collapse into a single whitespace. Text will wrap when necessary, and on line breaks

pre-wrap - Whitespace is preserved by the browser. Text will wrap when necessary, and on line breaks

WHITE-SPACE

```
p{white-space: nowrap;}
```

WORD-SPACING

The word-spacing property increases or decreases the white space between words. We can set this property to *normal* or specify in the form of length value.

```
p{word-spacing: 20px};
```

TEXT-ALIGN-LAST

The text-align-last property specifies how to align the last line of a text. The text-align-last property will only work for elements with the text-align property set to "justify". We can set this property to *auto*, *left*, *right*, *center*, *justify*, *start*, or *end*.

```
p { text-align: justify; text-align-last: right; }
```

WORD-WRAP

word-wrap property allows long words to be able to be broken and wrap onto the next line. We can set this property to either *normal* or *break-word*.

```
p { word-wrap: break-word; }
```

WORD-BREAK

The word-break property specifies line breaking rules for non-CJK scripts. CJK scripts are Chinese, Japanese and Korean ("CJK") scripts. We can set this property to *normal*, *break-all* or *keep-all*.

normal - Default value. Break words according to their usual rules

break-all - Lines may break between any two letters

keep-all - Breaks are prohibited between pairs of letters

```
p{word-break: break-all;}
```

TEXT-OVERFLOW

The text-overflow property specifies how overflowed content that is not displayed should be signaled to the user. We can set this property to *clip*, *ellipsis*, or *string*.

clip - Clips the text.

ellipsis - Render an ellipsis ("...") to represent clipped text

string - Render the given string to represent clipped text

```
p { text-overflow: ellipsis;}
```

FONT

The Font properties is used to set font related styles for the text present on an HTML.

- font-family
- font-size
- font-stretch
- font-style
- font-variant
- font-weight
- font

FONT-FAMILY

We can specify a list of font names for the text contained inside an HTML element, using font-family property.

Ex:-

```
p { font-family: Arial, serif, "Times New Roman"; }
```

If the name of a font family is more than one word, it must be in quotation marks.

Ex:-

“Times New Roman”

Times

FONT-SIZE

We can specify a font size for the text. We can specify size in the form of percentage, px, em.

Ex:-

```
p { font-size: 50%; }
```

```
p { font-size: 16px }
```

```
p { font-size: 2em; }
```

```
p { font-size: xx-small; }
```

- xx-small
- x-small
- small
- medium
- large
- x-large
- xx-large
- smaller
- larger

The default text size in browsers is 16px or 1em.

FONT-STRETCH

It specifies a normal, condensed or extended font face. We can set this property to *normal, wider, narrower, ultra-condensed, extra-condensed, condensed, semi-condensed, semi-expanded, expanded, extra-expanded, ultra-expanded*.

Ex:-

```
p { font-stretch: condensed; }
```

FONT-STYLE

It specifies the style of the font. We can set this property to *normal*, *italic*, *oblique*.

Ex:-

```
p { font-style: italic; }
```

By default text is normal

FONT-VARIANT

It specifies whether or not a font is a small-caps font. We can set this property to either *normal* or *small-caps*.

Ex: -

```
p { font-variant: small-caps; }
```

FONT-WEIGHT

The font weight property allows to set the font weight of the text present on an HTML Page. We can set this property to *normal*, *bold*, *bolder*, *lighter*, *number* (100, 200, 300 upto 900)

Ex: -

```
p { font-weight: bold; }
```

```
p { font-weight: 500; }
```

FONT

It defines a shorthand property for font-style, font-variant, font-weight, font-size, line-height, and font-family properties.

Ex:-

```
p { font: bold 20px serif;}
```

The properties must be set in order: "font-style font-variant font-weight font-size/line-height font-family"

The font-size and font-family values are required.

LIST

List property is used to create list. There are two type of lists:

- ▶ Unordered Lists – An unordered list uses bullets as the list item markers.
- ▶ Ordered Lists – An ordered list uses numbers or letters as the list item markers.
 - list-style-type
 - list-style-image
 - list-style-position
 - list-style

LIST-STYLE-IMAGE

In this an image is used as the list item marker. We can set this property to none or web address of an image.

Ex: -

```
ul { list-style-image: url('book.png'); }
```

LIST-STYLE-TYPE

It specifies the appearance of the list item maker if the **list-style-image** property has the value *none* or if the image specified through this property cannot be displayed. We can set this property to *none*, *disc*, *circle*, *square*, *decimal*, *decimal-leading-zero*, *lower-roman*, *upper-roman*, *lower-alpha*, *upper-alpha*, *lower-greek*, *lower-latin*, *upper-latin*, *hebrew*, *armenian*, *georgian*, *cjk-ideographic*, *hiragana*, *katakana*, *hiragana-iroha*, or *katakana-iroha*.

Ex: -

```
ul { list-style-type: circle; }
```

LIST-STYLE-POSITION

It specifies the position of the list item maker. You can set this property to either *inside* or *outside*.

Ex: -

```
ul { list-style-position: inside; }
```

LIST-STYLE

It is a shorthand property for *list-style-type*, *list-style-image* and *list-style-position*.

Ex:-

```
ul { list-style: square inside; }
```

Order: list-style-type, list-style-position, list-style-image.

BORDER-STYLE

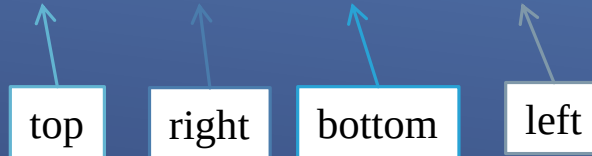
This property sets border style of an element. We can set this property to *none*, *hidden*, *dotted*, *dashed*, *solid*, *double*, *groove*, *ridge*, *inset*, *outset*.

Ex:-

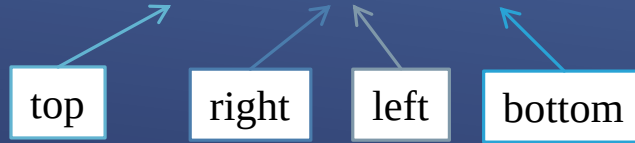
```
p { border-style: dashed; }
```

BORDER-STYLE

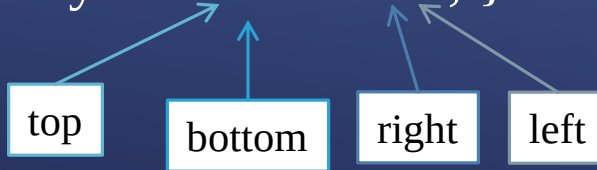
p { border-style: dotted solid double dashed; }



p { border-style: dotted solid dashed; }



p { border-style: dotted solid; }



p { border-style: dotted; }



BORDER-WIDTH

It sets width of element's border. We can set this property to thin, medium, thick, length in the form of px.

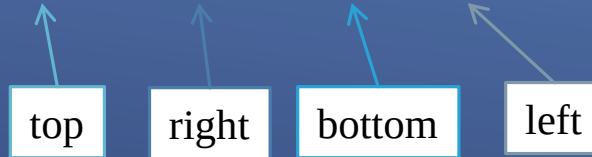
Ex:-

```
p { border-style: solid; border-width: thick; }
```

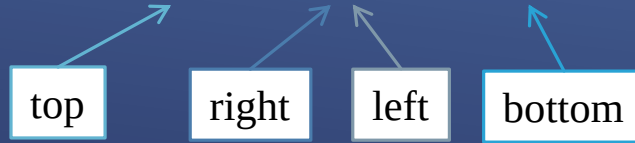
```
p { border-style: solid; border-width: 5px; }
```

BORDER-WIDTH

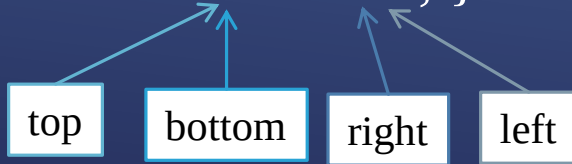
p { border-width: thick medium thin 5px; }



p { border-width: thick medium 5px; }



p { border-width: thick thin; }



p { border-width: 5px; }



BORDER-COLOR

It sets color of border. We can set this property to transparent and color value.

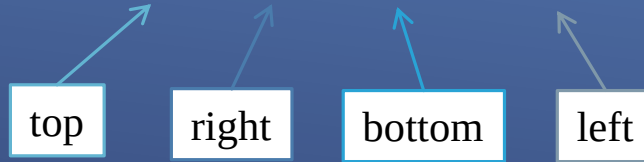
Ex:-

```
p { border-style: solid; border-color: blue;}
```

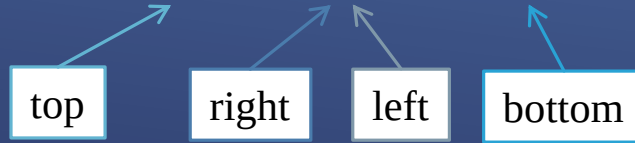
```
p { border-style: solid; border-color: #5A4218;}
```

BORDER-COLOR

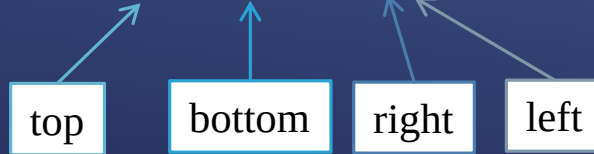
```
p { border-color: blue red black green; }
```



```
p { border-color: blue orange green; }
```



```
p { border-color: green orange; }
```



```
p { border-color: blue; }
```



BORDER

It is shorthand of all border properties. The order should be : border-width, border-style, and border-color.

Ex: -

```
p { border: 10px solid blue; }
```

```
p { border: solid blue; }
```

TABLE

- ▶ border
- ▶ border-collapse
- ▶ border-spacing
- ▶ caption-side
- ▶ empty-cells
- ▶ table-layout

BORDER

For specifying table we use border property.

Ex: -

```
table, th, td { border: 2px solid blue; }
```

BORDER-COLLAPSE

This property sets whether the table borders are collapsed into a single border or detached. We can set this property to *separate* (default) and *collapse*.

Ex:-

```
table { border-collapse: collapse; }
```

BORDER-SPACING

This property is used to set the distance between the borders of adjacent cells. This will work only when border-collapse is separated. We can set this property to horizontal spacing and vertical spacing in the form of cm, px.

Ex:-

```
table { border-collapse: separated; border-spacing: 5px 10px; }
```

CAPTION-SIDE

This property is used to place a caption for table. We can set this property to *top* (default) or *bottom*.

Ex:-

```
caption { caption-side: bottom; }
```

EMPTY-CELLS

This property is used to set whether or not to display borders and background on empty cells in a table. It will work only if the border-collapse is set to separate. We can set this property to *show* (default) or *hide*.

Ex:-

```
table { empty-cells: hide; }
```

TABLE-LAYOUT

This property is used to set table layout. We can set this property either *auto* (default) or *fixed* .

Ex: -

```
table { table-layout: fixed; }
```

HOW TO SET TABLE WIDTH AND HEIGHT

We use *width* and *height* property to set table width and height.

We can also set table heading (th) width and height.

```
table { width: 100% ;}
```

```
th { height: 10px; }
```

HOW TO SET TABLE PADDING

This property is used to control the space between the border and the content in a table, this is used on `<td>` and `<th>` elements.

```
th, td { padding: 10px; }
```

HOW TO SET TABLE COLOR

```
table {background-color: pink;}
```

DISPLAY

This property is used to define how an element should display. Every HTML element has a default display value depending on what type of element it is. The default display value for most elements is block or inline.

- inline
- block
- flex
- inline-block
- inline-flex
- inline-table
- list-item
- run-in
- table
- table-row-group
- table-caption
- table-column-group
- table-header-group
- table-footer-group
- table-cell
- table-column
- table-row
- none

DISPLAY

Block-level Elements - A Block-level element always starts on a new line and takes up the full width available.

Ex: -

<div>

<h1>

<p>

Inline Elements - An inline element does not start on a new line and only takes up as much width as necessary.

Ex: -

<a>

DISPLAY

- ▶ inline
- ▶ block
- ▶ flex
- ▶ inline-block
- ▶ inline-flex
- ▶ inline-table
- ▶ list-item
- ▶ run-in
- ▶ table

- ▶ table-row-group
- ▶ table-caption
- ▶ table-column-group
- ▶ table-header-group
- ▶ table-footer-group
- ▶ table-cell
- ▶ table-column
- ▶ table-row
- ▶ none

p { display: inline; }

DISPLAY

- ▶ inline – When we set this value, the element does not start on a new line and only takes up as much width as necessary (we can't set width/height it won't work)
- ▶ block – When we set this value, element always starts on a new line and takes up the full width available (we can set width/height)
- ▶ inline-block – It is combination of inline and block value. It doesn't start on new line but we can set width and height.
- ▶ none - The element will not be displayed at all (has no effect on layout)
- ▶ flex - Displays an element as a block-level flex container.
- ▶ inline-flex - Displays an element as an inline-level flex container.
- ▶ inline-table - The element is displayed as an inline-level table
- ▶ run-in - Displays an element as either block or inline, depending on context

DISPLAY

- ▶ table – It works like a `<table>` element
- ▶ table-caption - It works like a `<caption>` element
- ▶ table-column-group - It works like a `<colgroup>` element
- ▶ table-header-group - It works like a `<thead>` element
- ▶ table-footer-group - It works like a `<tfoot>` element
- ▶ table-row-group - It works like a `<tbody>` element
- ▶ table-cell - It works like a `<td>` element
- ▶ table-column - It works like a `<col>` element
- ▶ table-row - It works like a `<tr>` element
- ▶ list-item - It works like a `<li>` element

VISIBILITY

This property is used to specify whether or not an element is visible. We can set this property to visible (default), hidden or collapse.

- ▶ hidden - The element is invisible not removed.
- ▶ collapse – This value is only used for table elements. collapse removes a row or column, but it does not affect the table layout. The space taken up by the row or column will be available for other content. If collapse is used on other elements, it will be treated as “hidden”.

Ex:-

```
p { visibility: hidden;}
```

LINK

- ▶ a:link - a normal, unvisited link
- ▶ a:visited - a link the user has visited
- ▶ a:hover - a link when the user mouse over it
- ▶ a:active - a link the moment it is clicked

Ex:-

```
a:link {  
    color: red;  
}
```

a:hover MUST come after a:link and a:visited

a:active MUST come after a:hover

- ▶ left
- ▶ right
- ▶ top
- ▶ bottom



Auto
Length (cm, px, %)

left: auto
left: 55px
left: 50%

right: auto
right: 23px
right: 50%

top: auto
top: 20%
top: 40px

bottom: auto
bottom: 23cm
bottom: 40%

These properties are used to position an element.

We can not use any of these properties without setting up position property.

POSITION

▶ static

▶ fixed

▶ relative

▶ absolute

▶ left

▶ right

▶ top

▶ bottom

STATIC

When we set static position for an element then it is not positioned in any special way, it is always positioned according to the normal flow of the page.

left, right, top and bottom property won't work with static position.

Ex: -

```
h1 {  
    position: static;  
    border: 5px solid red;  
}  
  
h1 {  
    position: static;  
    border: 5px solid red;  
    left: 50px;  
}
```

FIXED

When we set fixed position for an element it always stays in the same place even if the page is scrolled. The top, right, bottom, and left properties are used to position the element.

```
img {  
    position: fixed;  
    right: 0;  
    top: 50%;  
    border: 3px solid red;  
}
```

RELATIVE

When we set relative position for an element, elements positioned to its normal flow.

```
h1 {  
    position: relative;  
    left: 50px;  
    border: 3px solid red;  
}
```

ABSOLUTE

When we set absolute position for an element, element is positioned to the nearest positioned ancestor. If there is no positioned ancestor then it follow the normal position according to browser.

```
div {  
    position: relative;  
    width: 400px;  
    height: 200px;  
    border: 3px solid red;  
}
```

```
div {  
    position: absolute;  
    top: 80px;  
    right: 0;  
    width: 200px;  
    height: 100px;  
    border: 3px solid red;  
}
```

HOW TO SET ELEMENT LAYER (Z-INDEX)

z-index property is used to set the layer of elements. We can set this property to *auto* (default) and *number*. It works only with position property value: fixed, relative, absolute.

The element which has higher number appear at the top.

Ex: -

```
img {  
    position: fixed;  
    z-index: 565;  
}
```